



ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANIZATION OF ISLAMIC CO-OPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING (EEE)



PROJECT REPORT

PROJECT TITLE: Machine learning-based Crop Recommendation System

Course Name: Artificial Intelligence And Machine Learning Lab

Course Code: EEE 4710

Section: A

NAME

Bah Alusine

ID

210021165

Abstract

One of the sectors that can be important in regard to the sustenance of human life is agriculture. The choice of the appropriate crop depending on the soil and environmental factors is crucial to enhancing the productivity and reducing the wastage of resources. The given project proposes a machine learning-oriented Crop Recommendation System, which is able to determine the most appropriate crop based on such parameters as the nutrients of the soil and the climatic conditions.

The system is trained with the data set comprising of such features as nitrogen, phosphorus, potassium, temperature, humidity, pH, rainfall. Several machine learning models were employed and tested and the most effective was chosen to predict. The Python environment was also used to create a simple user interface that enabled users to input parameters and get the crop recommendations immediately.

1. Introduction

The agricultural sector is an important part of the economy of most states. When making decisions on which crop to grow, farmers tend to use their preexisting knowledge and experience as a basis. Nevertheless, due to the shifting environment and the growing needs of food, there has been a need to incorporate modern methods of making decisions.

Machine learning is a good solution to analyze agricultural data and make correct predictions. The current project is devoted to the creation of a Crop Recommendation System that will be used to guide farmers by providing recommendations on the most appropriate crop with the input parameters provided.

2. Problem Statement

In crop selection, farmers have a number of challenges that include:

Poor knowledge on soil nutrients.

Uncertainty in weather conditions

Poor utilization of water and fertilizers.

Poor performance in terms of wrong choice of crops.

This project is targeted to design a system capable of analyzing soil and environmental data and give correct crop recommendations.

3. Objectives

The main objectives of this project are:

1. To create a machine learning model to predict crops.
2. To examine soil and climatic parameters.
3. To enhance agricultural decision-making.

4. Dataset Description

The data employed in this project has the agricultural parameters that determine the growth of crops. The features include:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- pH value
- Rainfall

The crop label is the target variable, and it is the most appropriate crop to be planted given the conditions.

All the records in the dataset represent a particular set of soil and environmental conditions and the recommended crop.

5. Methodology

The project was conducted in a well-organized way as follows:

5.1 Data Loading

The system was loaded with Python libraries like Pandas to load the dataset into the system.

5.2 Data Preprocessing

- Checked for missing values
- Verified data types
- Disconnected input values and target variable.
- StandardScaler feature scaling.

5.3 Feature Selection

The model was trained on the following features:

- Nutrients of soil (N, P, K)
- Conditions of the surroundings (temperature, humidity, rainfall).
- Soil pH

5.4 Model Training

A number of machine learning models were trained, such as:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors
- Support Vector Machine

5.5 Model Evaluation

The models were checked on the basis of accuracy. Final prediction was done on a model that had the best performance.

6. Model Selection

Upon the comparison of the different models, the best model was discovered to be the Random Forest Classifier. It provided:

High accuracy

Better generalization

Stability on the various inputs.

Thus, it was decided to choose Random Forest as the ultimate system model.

7. System Implementation

Python was used to implement the system in Google Colab. The steps involved:

- Camping and navigation of the dataset.

- Preprocessing the data
- Splitting data into training and testing sets
- Training multiple models
- Choosing the most appropriate model.
- Developing a prediction capability.

8. System Interface

The interface with a simple user interaction was provided in Python environment. The system enables users to manually feed in values and get predictions of crops instantly.

The user makes the following inputs:

- Nitrogen
- Phosphorus
- Potassium
- Temperature
- Humidity
- pH
- Rainfall

These values are processed by the trained machine learning model, which predicts the most suitable crop.

The interface is similar to a simple calculator and thus can be easily utilized without having to have a lot of technical understanding.

9. Results

The designed system is able to anticipate crops using the input parameters. The model is very accurate and generates dependable recommendations.

Example output:

Recommended Crop: Rice

The system is effective and offers fast predictions thus suitable in practical applications.

Crop Recommendation System

Enter soil and weather parameters to predict the best crop.

Nitrogen

70



Phosphorus

60

Potassium

49

Temperature

40

Humidity

30

pH

20

Rainfall

80

Clear

Submit

output

papaya

Flag

[Use via API](#)  · [Built with Gradio](#)  · [Settings](#) 

10. Advantages

- Easy to use
- Swift and speedy forecasts.
- Assists in making decisions.
- Eliminates reliance on old-fashioned guesswork.
- Improves agricultural productivity

11. Limitations

- Relies on the quality of dataset.
- Lacks weather updates are not real-time.
- Restricted to the characteristics in the dataset.

12. Future Improvements

Enhancements in the system can be made in future by:

1. Adding live weather information.
2. Creation of a web or mobile application.
3. Increasing the data to enhance accuracy.
4. Observing more advanced machine learning/deep learning models.
5. Adding support for multiple languages

13. Tools and Technologies Used.

- Python
- Google Colab
- Pandas and NumPy
- Scikit-learn
- Matplotlib and Seaborn

14. Conclusion

Crop Recommendation System shows how machine learning can be efficiently applied in agriculture. Through the study of the soil nutrients and environmental conditions, the system gives precise crop recommendations. This method can assist the farmers to make better decisions, to increase crop production and also to lessen waste of resources. The project outlines the potential of technology in addressing actual agricultural issues and in promoting sustainable agriculture.